Course report
SF2321 Applied Statistical Analysis
Fall 2022
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Introduction

The course SF2321 Applied Statistical Analysis is a second-cycle method course for MA students in Criminology[1], European Studies (MAES), International Administration and Global Governance (IAGG), Political Communication (POLCOM), Political Science (MAPS) , Political Science: Environmental Governance and Behavior (MAPSE) and Sociology.[2] The 2021 edition was the fourth time the course was held in its current form.[3]

Five teachers, all associate professors, from the Department of political science taught on the 2021 course edition: Mattias Agerberg Rasmus Broms, Magnus Lundgren, Maria Solevid and Aksel Sundström. Mattias joined the course to replace Nicholas Charron who was on sabbatical. As usual, all teachers were involved in supervision and examination of the method theses

In 2022, 41 students were registered on the course, 36 first-time registered and 5 re-registered. 37 students, all of the first-time registered and one of the re-registered students were active and took credits. Of the 36 first-time registered and active students, seven was from IAGG, three from MAES, seven from MAPS, nine from MAPSE, two from POLCOM, five from Sociology and three free-standing students.

The course was taught on campus with the addition of some well-thought through digital teaching events. We have deliberately kept the pre-recorded introduction to the Stata tutorials, where the instructor demonstrates the code and procedures in Stata relevant for the assignment (1-5). The decision to keep these pre-recorded tutorials was initially based on the feedback from the 2020 student cohort, but the recorded tutorials was also very appreciated among the 2021 cohort as well. The previous student cohorts have expressed their appreciation of these tutorials both in the course evaluation survey as well as in direct discussion with the course coordinator during the MAPS/MAPSE education program day. The decision to keep the video tutorials also means that the students get one hour more hands on help each week since the time otherwise spent on demonstrating the commands live in the classroom instead could be spent on helping the students. We also carry out the two advice

[1] MA students in Criminology participate in one of the two quantitative method courses every other year (2019, 2021 and so on).

[2] The course started in 2007 and was then a formal collaboration between Department of Journalism, Media and Communication, Department of Sociology and Work Science and Department of Political Science (coordinating department), but the participating MA programs has shifted over the years. Throughout the years, the course has been revised several times. Before 2018, the largest revision took place in 2012 were both learning outcomes were revised and the teaching language changed from Swedish to English.

[3] The course revisions were part of a large method course block revision initiated by the department of Political Science. As a result of this revision, the students from the three MA programs with the department (IAGG, MAES and MAPS) can now choose between the same set of method courses: two qualitative courses with different method emphasis (one coordinated by Sociology and Work Science) and two quantitative method courses, one more basic course for students without prior training in statistical methods and one more advanced course for students with prior training in statistical methods. SF2321 Applied Statistical Analysis is the more advanced method course in statistics. In terms of learning outcomes, they emphasize advanced regression techniques and also explicitly mentions all the regression techniques the course now covers. Current syllabus can be found at http://kursplaner.gu.se/english/SF2321.pdf.

sessions related to the Method thesis on Zoom, this as these sessions take place just before, and in the middle of, the Christmas and New Years holidays. The learning from the pandemic is that supervision/thesis advice session works very good in a digital format too. Content-wise, we added one new lecture, on the topic causal inference.

Course structure

SF2321 Applied Statistical Analysis is a course where theory-based statistical analysis is taught through different regression methods. Table 1 below summarizes the course structure of the 2022 course, and the structure is the same as in the 2018–2021 course editions.

The course starts with a Recap section with one lecture on statistical inference and one lecture on aspects such as variable language, data matrix, variable/variable units, descriptive statistics, cross-tabs and correlations and tutorial with introduction to Stata.

Table 1 Overview of course structure, 2022

	Course structure 2019	Responsible teacher
Week 1	Recap and introduction to Stata	Maria Solevid/Aksel Sundström
Week 2	Introduction to linear regression (part 1)	Aksel Sundström
Week 3	Introduction to linear regression part 2 (part 1+part 2=3 hec)	Aksel Sundström
Week 4	Statistical elaboration with linear regression (1,5 hec)	Maria Solevid
Week 5	Introduction to categorical regression (1,5 hec)	Magnus Lundgren
Week 6	Introduction to multi-level regression (1,5 hec)	Mattias Agerberg
Week 7	Introduction to panel regression (1,5 hec)	Rasmus Broms
Week 8-10	Method thesis (6 hec)	Maria Solevid (all teachers involved)

After the introductory part, the first workshop is Introduction to regression. This workshop spans over two weeks and covers both the basics of linear regression as well as the diagnostics. The regression workshop is followed by yet another week on linear regression, but this time on statistical elaboration through mediation and moderation. In this workshop, designs for causal identification are also brought up. After these three weeks of linear regression, three weeks follows in which one more advanced regression technique is introduced every week: Categorical regression, Multi-level regression and Panel regression. The course ends with an independent research paper, the method thesis. The five workshops are 9 hec (each is 3 or 1,5 hec) and the method thesis is 6 hec.

Completion rate

The completion rate at the end of the course (including the revision round for those that didn't reach pass, but also didn't receive fail) was 63 percent among all registered students, including the re-registered students and 69 percent among first-time registered students. The completion rate among all students is lower compared to 2021 (69 percent) and 2018 (67 percent) but higher compared to 2020 (53 percent) and 2019 (58 percent). Previous years, completion rates have varied a lot depending on number of active students – some years, more students who register are also active, other years fewer registered students are active. The completion rate among the 37 active students in 2022 is 70 percent which is lower compared to the years for which we have a comparable measure; 2021 (90 percent), 2020 (76 percent), and 2019 (76 percent).

Of the 26 students who had completed the course during the original submission rounds, six were awarded Pass with distinction and 20 Pass. The share of Pass with distinction is 23 percent which is on par with 2021 (22 percent) but lower compared to the previous years (2020: 31%, 2019: 42%, 2018: 38%).

Students' assessment of core course elements

Of the 41 registered students, 22 answered the course evaluation survey which equals a response rate of 53,7 percent[4]. However, if we consider that 37 students have been active throughout the course, a fairer response rate is probably 59,5 percent. In addition to scheduling time for the course evaluation at the end of the final seminar, invitation and reminders were sent out through the SUNET survey system, and additional reminders were made by the course coordinator through Canvas messages.

The almost 54 percent response rate is an increase compared 2021 and 2020, but lower compared to 2019 which was the last the course evaluation was scheduled at the end of the final seminar.

Table 2 summarizes the students' assessment of the course in general, the lectures, teachers, computer labs and overall course organization/administration. Most tables present both percentages and means and for comparison, the means from 2018–2021 are also displayed.

Table 2 General assessments of the course, lectures, tutorials, teachers and course administration (percent and mean)

	The course in general	Lectures	Stata tutorials	Teachers	Organisation and administration of the course
Very good	27	14	27	27	55
Rather good	50	64	41	55	41
Neither bad nor good	9	18	32	18	4
Rather bad	14	4	0	0	0
Very bad	0	0	0	0	0
Total	100	100	100	100	100
N	22	22	22	22	22
Mean 2022	**3,9**	**3,9**	**4,0**	**4,1**	**4,5**
Mean 2021	**4,4**	**4,1**	**4,4**	**4,6**	**4,5**
Mean 2020	**4,4**	**4,0**	**4,1**	**4,6**	**4,6**
Mean 2019	**4,3**	**3,8**	**4,3**	**4,1**	**4,4**
Mean 2018	**4,1**	**3,9**	**4,0**	**3,9**	**4,4**

Comment: Question formulations: "Overall, how would you assess the….?". Very bad coded as 1 and Very good coded as 5. For the question on distance teaching, the question wording is "In general, how do you think the distance teaching worked?".

[4] In 2020 and 2021, we verbally reminded the students during the Zoom seminar to fill out the course evaluation survey after the seminar. RR 2021: 46 percent of all/61 percent among active, RR 2020: 53 percent among all/76 percent among active. In 2019, with scheduled time for the course evaluation in the classroom, the response rate was 70 percent among all registered students. In 2018, the response rate was 59 percent without scheduled time to fill out the course evaluation survey..

The core course elements receive overall positive assessments. 77 percent of the students answers that the course in general is rather or very good. The mean on a 1-5 scale, were 1 equals very bad and 5 equals very good, is 3,9 and unfortunately the lowest score we have seen. This is a bit surprising to us as the course set up has not changed. 78 percent respond that the lectures are rather or very good (mean 3,9), 68 percent shares the same view of the Stata tutorials (mean 4,0) and 82 percent make positive assessments of the teachers (mean 4,1). 96 percent assesses the organization and administration of the course as rather or very good (mean 4,5). Overall, the 2022 average assessments are lower compared to previous years. As no major changes have been made over the years, only incremental improvements, we need to think more carefully about how we inform students about what knowledge they need beforehand. The lower completion rates and the slightly less positive assessments are two indicators of that the students might have struggled more with the course.

Students' assessment of the different workshops

Table 3 below summarizes students' assessments of the different workshops throughout the course. The general conclusion is that all workshops receive overall positive evaluations with average assessment ranging from 3,7 to 4,2. Once again, it is evident that the 2022 cohort on average is less positive compared to the 2021 cohort (workshop ratings in 2021 ranged from 3,9 to 4,8). If we focus on the workshops connected to assignment 1-5, 72 percent of the responding students assesses Introduction to linear regression as very or rather good. The equivalent numbers for the other workshops are 81 percent (statistical elaboration regression), 81 percent (categorical regression), 60 percent (multilevel regression) and 76 percent (panel regression). Compared to 2021, statistical elaboration and multi-level regression see the largest drops in degree of positive assessments, however these two workshops showed very high averages in 2021 (4,8)

The Method thesis receives an average assessment of 3,8 (74 percent positive approvals) which on par with 2021 and 2019, but lower compared to 2020.

Table 3 Evaluations of the different workshops (percent and mean)

	Recap and introduction to Stata	Introduction to linear regression	Statistical elaboration with linear regression	Introduction to categorical regression	Introduction to multi-level regression	Introduction to panel regression	Method thesis
Very good	14	24	43	48	20	33	16
Rather good	48	48	38	33	40	43	58
Neither bad nor good	29	14	9	3	25	19	21
Rather bad	9	14	5	0	3	5	5
Very bad	0	0	5	1	0	0	0
Total	100	100	100	100	100	100	100
N	21	21	21	21	20	21	19
Mean 2022	**3,7**	**3,8**	**4,1**	**4,2**	**3,7**	**4,0**	**3,8**
Mean 2021	**4,2**	**4,0**	**4,8**	**4,5**	**4,6**	**4,1**	**3,9**
Mean 2020	**4,0**	**4,3**	**4,3**	**4,1**	**4,4**	**4,2**	**4,5**
Mean 2019	**3,9**	**4,7**	**4,5**	**2,9**	**2,7**	**4,3**	**3,8**
Mean 2018	**3,7**	**4,4**	**4,5**	**2,7**	**2,9**	**4,5**	**4,1**

Comment: Question formulations: "What is your assessment of the different workshops specified below?" Very bad coded as 1 and Very good coded as 5. Recap and introduction to Stata is the only workshop not associated with an assignment. Linear regression is assignment 1, Method thesis is assignment 6.

Students' assessments of examination forms, literature, amount of work and Eureka moment.

As shown in table 4, the examination forms – the five workshop assignments and the method thesis respectively – receives average assessments of 4,0 and 3,7 respectively. 73 and 64 percent of the students respond that these examination forms are rather good or very good. It is a notable that the method thesis as examination form is viewed less positive in 2022 compared to previous years.

Table 4 Evaluation of the examination forms

	Workshops assignments as examination forms	Method thesis as examination form
Very good	23	9
Rather good	50	55
Neither bad nor good	27	32
Rather bad	0	4
Very bad	0	0
Total	100	100
N	22	22
Mean 2022	**4,0**	**3,7**
Mean 2021	**4,2**	**4,5**
Mean 2020	**4,2**	**4,3**
Mean 2019	**4,2**	**4,2**
Mean 2018	**3,8**	**4,2**

Comment: Question formulation: Overall, how would you assess the examinations? Very bad coded as 1 and Very good coded as 5

Table 5 Evaluation of the literature

	Aneshensel Theory-based data analysis for the social sciences	Mehmetoglu & Jakobsen Applied statistics using Stata
Very good	4,5	27
Rather good	36	59
Neither bad nor good	32	9
Rather bad	23	5
Very bad	4,5	0
Total	100	100
N	23	23
Mean 2022	**3,1**	**4,1**
Mean 2021	**3,6**	**4,7**
Mean 2020	**3,4**	**4,4**
Mean 2019	**3,5**	**4,4**
Mean 2018	**3,8**	**4,5**

Comment: Question formulation: "Overall, how would you assess the literature?" followed by the two author names and titles of the two course books. Very bad coded as 1, Very good coded as 5. In addition to the response options displayed in the table, the option "I didn't read this book" was also available but has been excluded from the mean calculations.

The course literature consists of two textbooks; the "theory book" is *Theory-based data analysis for the social sciences* by Carol Aneshensel and the "applied statistics book" is *Applied statistics using Stata* by Mehmet Mehmetoglu and Tor Jakobsen. In short, the book by Mehmetoglu and Jakobsen receives an average assessment of 4,1 on the 1 to 5 scale where 1 is very bad and 5 is very good (see table 5). This is the least positive assessment since we introduced this book in 2018. The average assessment for the book by Aneshensel is 3,1 which also is the lowest rating.

The assessment regarding amount of work and how many hours the student spent on course work (inside and outside the classroom) is reported in table 6. The results show firstly that 18 percent says the workload was about right (in 2021 this figure was 48 percent), but most notablu that 73 percent think it was somewhat too demanding or too demanding. 41 (in 2021: 20 percent) of the students answers they spent more than 40 hours per week on the course work, 46 percent answers 31-40 hours, 4 percent 21-30 hours and 9 percent 11-20 hours. Judging from the displayed averages, the course was viewed as more demanding compared to previous years, and students also reports spending a lot of time on course work. As the course content is identical to 2021, and only very minor changes – improvements based on comments from previous cohorts – have taken place since 2018, we once again reach the conclusion that the students likely had less previous knowledge or experience than what we expect them too.

Table 6 Evaluation of amount of work

Amount of work		How much time spent on course work	
Too demanding	27	More than 40 hours	41
Somewhat too demanding	55	31-40 hours	46
About right	18	21-30 hours	4
Not that demanding	0	11-20 hours	9
Not at all demanding	0	Less than 10 hours	0
Total	100	Total	100
N	22	N	22
Mean 2022	4,1	Mean 2022	4,2
Mean 2021	3,6	Mean 2021	3,9
Mean 2020	3,4	Mean 2020	3,4
Mean 2019	3,5	Mean 2019	4
Mean 2018	3,7	Mean 2018	4,1

Comment: "What is your assessment of the amount of work you have been requested to carry out during the course?". Not at all demanding coded as 1, Too demanding coded as 5 "On average, approximately how much time did you spend on coursework, both inside and outside the classroom, per week?". Less than 10 hours coded as 1, More than 40 hours coded as 5. The correlation (Pearson's R) between the two questions is .46.

59 percent of the students answer that they experienced a Eureka moment during the course (not shown in table), this is unfortunately lower compared to 2021 and 2020 (72/96 percent).

Students' expectations and learning process

In the discussion and research about course evaluations, there is a drive towards a transition from pure product assessments (how satisfied the students are with the 'delivery' of the course or the 'performance' of teacher) to assessments based on students' own learning process. This implies a shift from quality control to quality development. As stated by Zerihun et al (2012:102), "students will be in a better position to provide appropriate feedback for teaching improvement if the evaluation of teaching quality also addresses their learning experiences".[5]

To facilitate student reflections about their own learning, I initiated beehive discussions at the course introduction lecture. The students were asked to in small groups discuss two questions: 1) What do you expect (or want/wish) to learn during the course? 2) How do you plan to achieve the expectations regarding your own learning? and to write down their answers and provide them to me.

To follow-up this introductory group discussion, I included a text in the course evaluation with the intent to remind students about these initial discussions, followed by a question of expectation fulfilment. 38 percent of the students responded that their expectations were fulfilled, 29 percent that expectations were more than fulfilled, 24 percent that expectations were partially fulfilled and 9 percent that the expectations were not fulfilled. (see table 7). In comparison to previous years, the average assessment is on par with 2019, but lower compared to both 2020 and 2021.[6]

Table 7 **If you think back to your initial reflections and discussion about your own learning expectations of the course, to what extent would you say your expectations have been fulfilled? (percent and mean)**

Expectations more than fulfilled	29
Expectations fulfilled	38
Expectations partially fulfilled	24
Expectations not fulfilled	9
Sum	100
N	21
Mean 2022	2,9
Mean 2021	3,3
Mean 2020	3,1
Mean 2019*	2,8

Comment: Question reads: "At the course introduction, you were asked to in small groups reflect and discuss your learning expectations and how you, through your own learning, should achieve those expectations. If you think back to your initial reflections and discussion about your own learning expectations of the course, to what extent would you say your expectations have been fulfilled?" In addition to the response options displayed in the table, option "No opinion/I didn't attend the course introduction" was also offered but no

[5] Zerihun, Z, Beishuizen, J & Van Os, W. 2012. Student learning experience as indicator of teaching quality. *Educational Assessment, Evaluation and Accountability*, 24:99-111.

[6] In 2019, to test whether the priming on expectations yielded any effect on evaluations, I set up a simple experiment. The course evaluation survey contained identical questions, but the order of the expectation fulfilment question was randomized to appear either as the first or the last question. The hypothesis was that students who are reminded about the initial group discussions, and thereby primed to reflect about their own learning, at the start of the survey should make more positive assessments of the course compared to students receiving this question at the end of the survey. The result showed a slightly higher mean in the first group, indicating more positive assessments in terms of expectation fulfilment, but the mean difference was too small to yield a substantial effect and the difference didn't reach statistical significance. When comparing course assessments between the two groups, students answering the reflection question first made slightly more positive assessments about the course in general, the lectures and the teachers compared to student answering the reflection question as the last survey question.

students answered this. Expectations not at all fulfilled coded as 1, Expectations more than fulfilled coded as 4.
*The 2019 average is the combined average from the two experiment groups

Reflections

Overall, we are satisfied with how the course went. This is also mostly reflected in the students' assessments which overall are on the positive side of the response scale. However, it is at the same time evident that the students are less positive compared to previous cohorts. As only very minor changes have taken place over the years, our working hypothesis is that the students were less prepared for the course compared to previous cohorts and one reason might be the nature of their previous methods training. The conclusion is based on that we saw slightly lower completion rates, and many students who repeatedly received revisions on their course work. More students also had to revise their method thesis and we also saw more theses in the March submission round, which is the first re-take exam possibility. It is also evident the 2022 cohort has worked very hard with over 40 percent estimating the course work took more than 40 hours per week and a majority who responded the course was slightly too demanding or too demanding. We acknowledge that the students' have worked hard, but we were nevertheless not prepared for that the shares of students working more than 40 hours per week was as high as it was. We also acknowledge that students perceive the workload as uneven during the workshop part of the course.

For some time, the team of teachers have been discussing ways to further develop the course, and it has now been decided to make changes to the course. The background to the identified development and change is that we want to bring the course content closer to the research and methods front in political science by introducing design and statistical methods for causal inference, that is how to be able to conclude that "x *causes* y". The lecture we added in 2022 was a first step. From 2023, the course will have revised learning outcomes and content. The course will start will lectures on statistical inference and research design for causal inference, including a small assignment on research design. After this introduction, we will hold the recap lecture and the introduction to Stata, followed by introduction to linear regression and heterogenous effects (interaction models) in linear regression. After these three weeks, we will devote three weeks to methods for causal inference: panel regression, difference-in-difference and synthetic control. The course will, just as today, end with a method thesis. Compared to the current version of the course, we will mainly focus on linear regression models, but increase the complexity in terms of design and data required to draw conclusions about causal effects.